# Explaining Models or Modelling Explanations

## Challenging Existing Paradigms in Trustworthy AI

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# Background

- Economist, now PhD CS
- How can we make opaque AI more trustworthy?
- Explainable AI, Adversarial ML, Probabilistic ML
- Maintainer of Taija (trustworthy AI in Julia)




Figure 1: Scan for slides.
Links to www.patalt.org.

# Agenda

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- What dynamics are generated when off-the-shelf solutions to CE and AR are implemented in practice?
- Can we generate plausible counterfactuals relying only on the opaque model itself?
- How can we leverage counterfactuals during training to build more trustworthy models?

# Background

# CE in Five Slides



Figure 2: Cats and dogs in two dimensions.

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**Objective**:

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$$\theta^* = \theta_T$$

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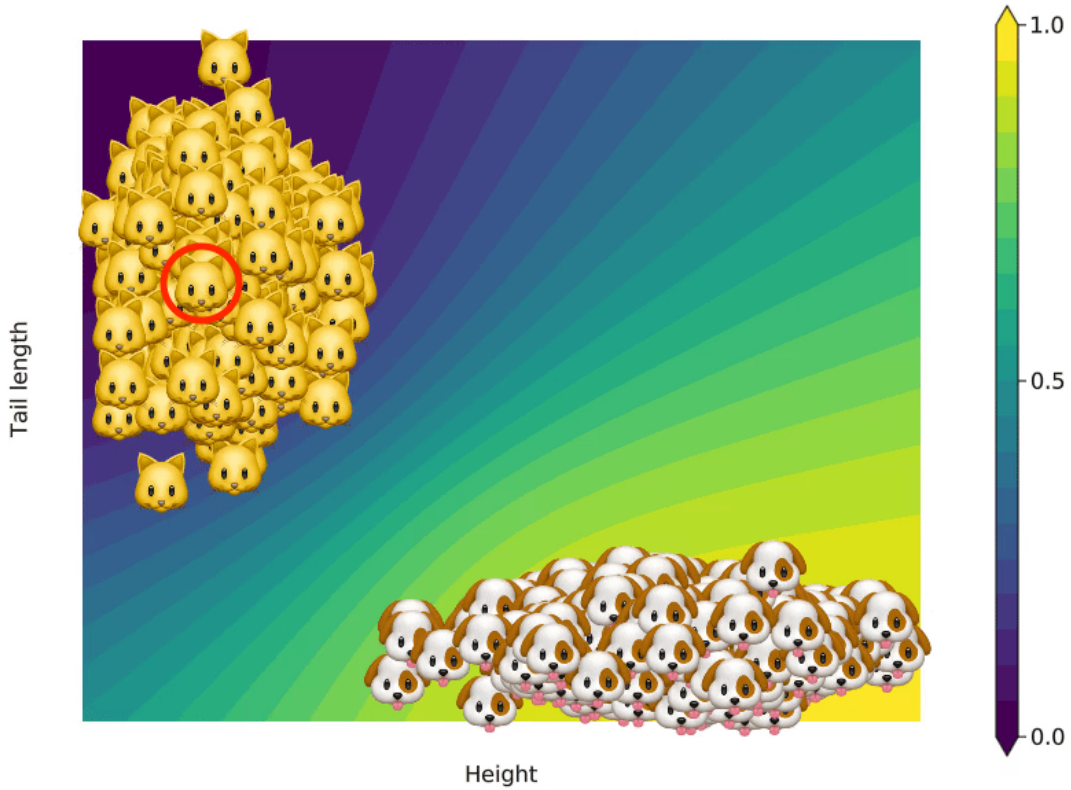



Figure 3: Fitted model. Contour shows predicted probability $y =$ .

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# CE in Five Slides

$$\min_{\mathbf{Z}' \in \mathcal{Z}^L} \{\text{yloss}(M_\theta(f(\mathbf{Z}')), \mathbf{y}^+) + \lambda \text{cost}(f(\mathbf{Z}'))\}$$

**Counterfactual Explanations** explain how inputs into a model need to change for it to produce different outputs[1].

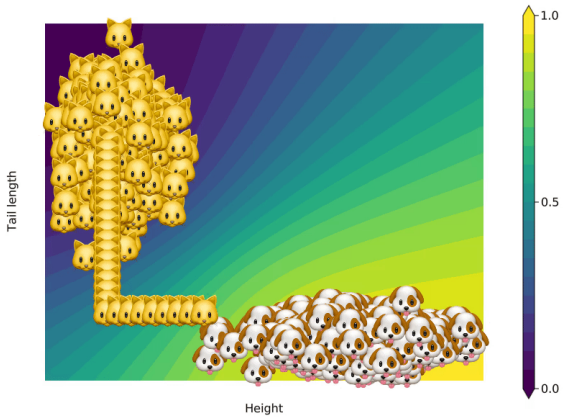



Figure 4: Counterfactual explanation for what it takes to be a dog.

[1] Altmeyer, Deursen, et al. (2023) @ JuliaCon 2022

# Algorithmic Recourse

Provided CE is valid, plausible and actionable, it can be used to provide recourse to individuals negatively affected by models.

> *"If your income had been X, then ..."*

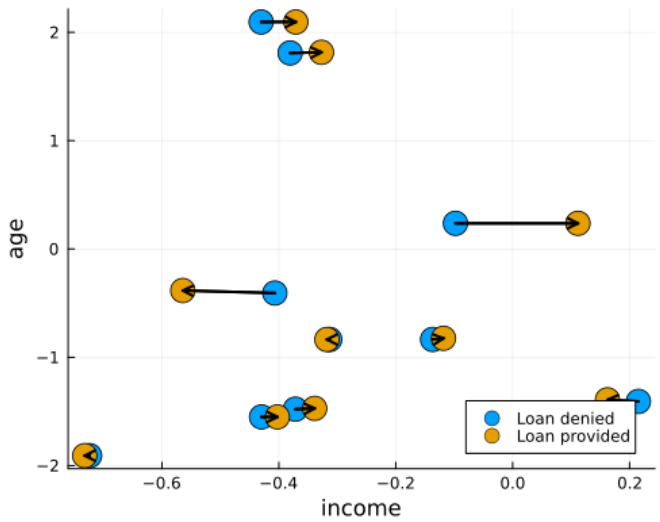



Figure 5: Counterfactuals for random samples from the Give Me Some Credit dataset (Kaggle 2011). Features 'age' and 'income' are shown.

# Dynamics of CE and AR

# Hidden Cost of Implausibility

AR can introduce costly dynamics[2]

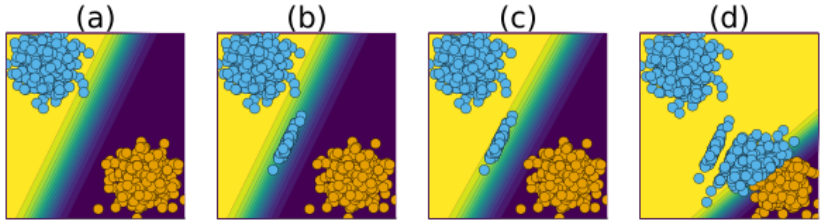



Figure 6: Endogenous Macrodynamics in Algorithmic Recourse.

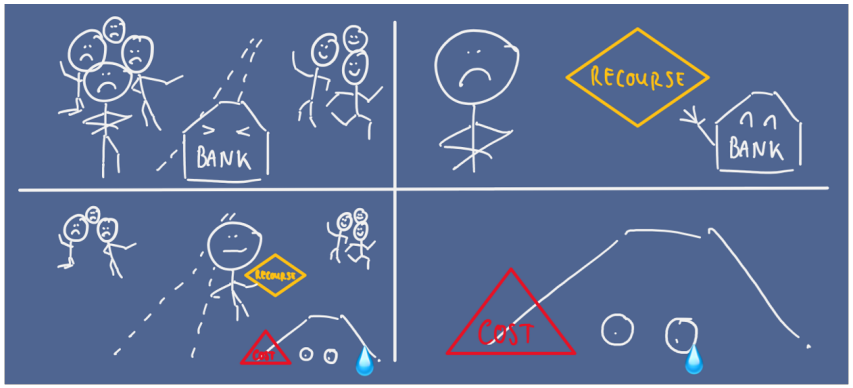



Figure 7: Illustration of external cost of individual recourse.

🔑 **Insight**: individual recourse neglects bigger picture.

[2] Altmeyer, Angela, et al. (2023) @ SaTML 2023.

# Mitigation Strategies

- Incorporate hidden cost in reframed objective.
- Reducing hidden cost is equivalent to ensuring plausibility.

$$\mathbf{s}' = \arg\min_{\mathbf{s}' \in \mathcal{S}} \{\text{yloss}(M(f(\mathbf{s}')), y^*) + \lambda_1 \text{cost}(f(\mathbf{s}')) + \lambda_2 \text{extcost}(f(\mathbf{s}'))\}$$

# Plausibility at all cost?

# Pick your Poison

All of these counterfactuals are valid explanations for the model's prediction.

*Which one would you pick?*

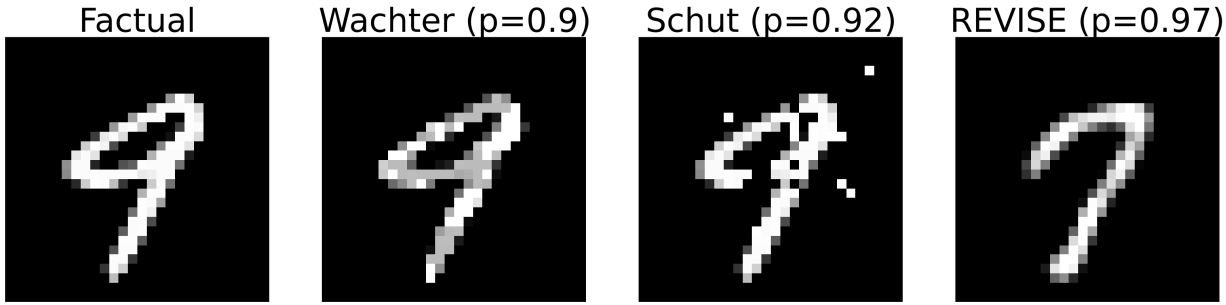



Figure 8: Turning a 9 into a 7: Counterfactual explanations for an image classifier produced using *Wachter* (Wachter et al. 2017), *Schut* (Schut et al. 2021) and *REVISE* (Joshi et al. 2019).

# Faithful First, Plausible Second

Counterfactuals as plausible as the model permits[3].

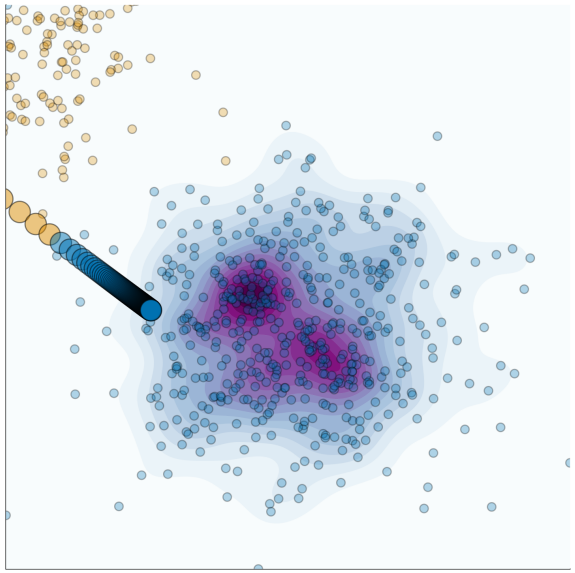


Figure 9: KDE for training data.

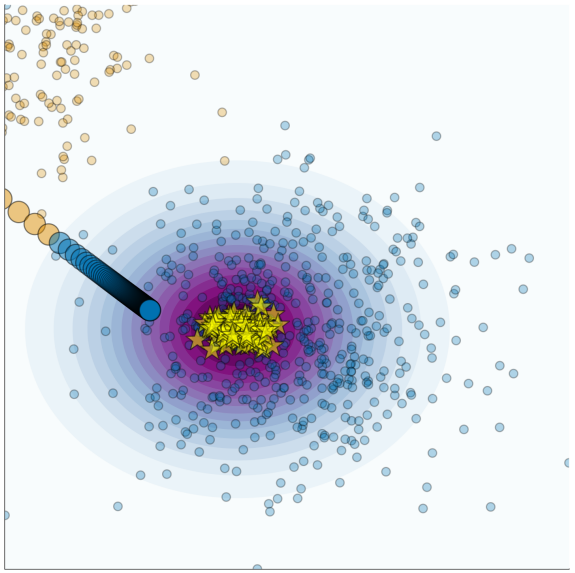


Figure 10: KDE for model posterior.

[3] Altmeyer, Farmanbar, et al. (2023) @ AAAI 2024. [blog]

# Faithful Counterfactuals

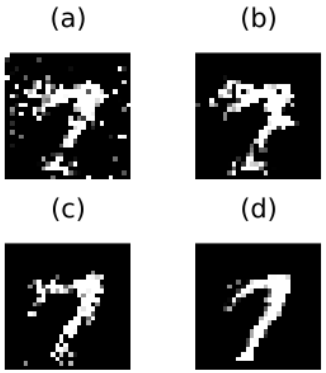



Figure 11: Turning a 9 into a 7. *ECCCo* applied to MLP (a), Ensemble (b), JEM (c), JEM Ensemble (d).

**Insight**: faithfulness facilitates

- model quality checks (Figure 11).
- state-of-the-art plausibility (Figure 12).

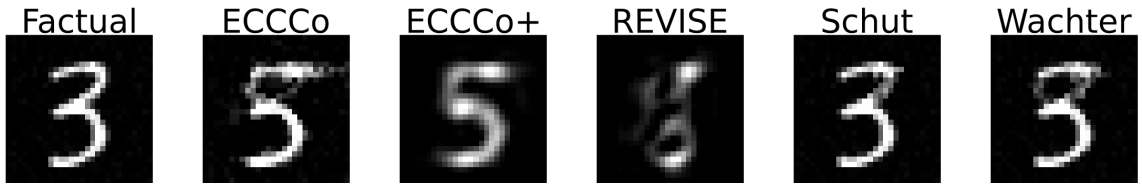



Figure 12: Results for different generators (from 3 to 5).

# Teaching models plausible explanations

# Counterfactual Training: Method

**Idea**

Let the model compare its own explanations to plausible ones[4].

1. Contrast faithful counterfactuals with data.
2. Use nascent CE as adversarial examples.

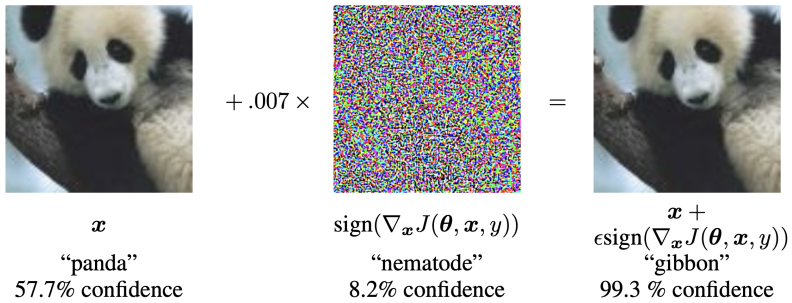



Figure 13: Example of an adversarial attack. Source: Goodfellow et al. (2015)

[4] under review

# Counterfactual Training: Results

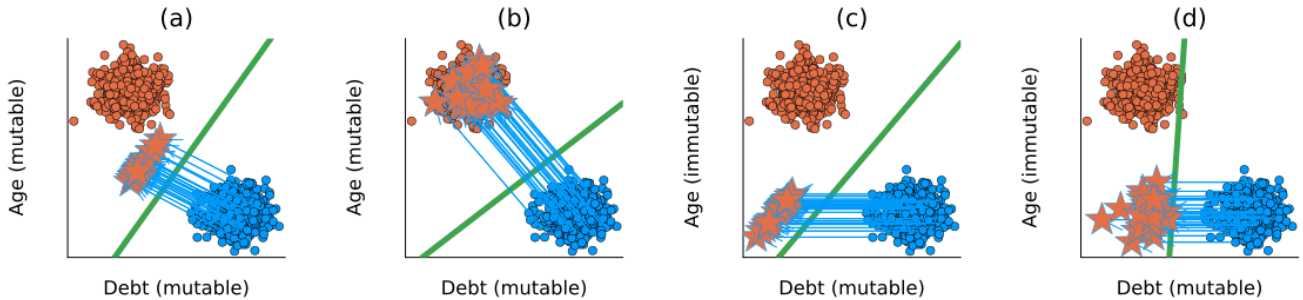



Figure 14: (a) conventional training, all mutable; (b) CT, all mutable; (c) conventional, *age* immutable; (d) CT, *age* immutable.

- Models trained with CT learn more plausible and (provably) actionable explanations.
- Predictive performance does not suffer, robust performance improves.

# If we still have time ...

# Spurious Sparks of AGI

We challenge the idea that the finding of meaningful patterns in latent spaces of large models is indicative of AGI[5].

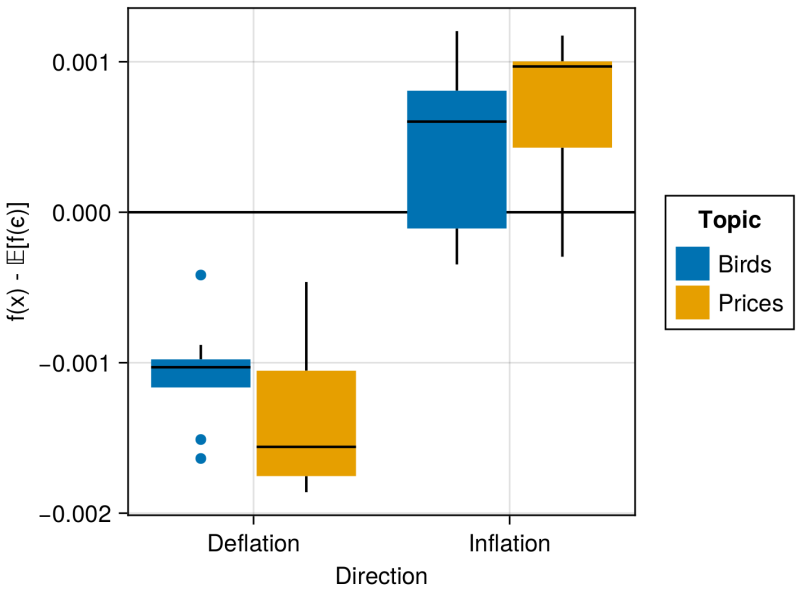



Figure 15: Inflation of prices or birds? It doesn't matter!

[5] In Altmeyer et al. (2024) @ ICML 2024

# Taija

- Model Explainability (CounterfactualExplanations.jl)
- Predictive Uncertainty Quantification (ConformalPrediction.jl)
- Effortless Bayesian Deep Learning (LaplaceRedux.jl)
- ... and more!

- Work presented @ JuliaCon 2022, 2023, 2024.
- Google Summer of Code and Julia Season of Contributions 2024.
- Total of three software projects @ TU Delft.

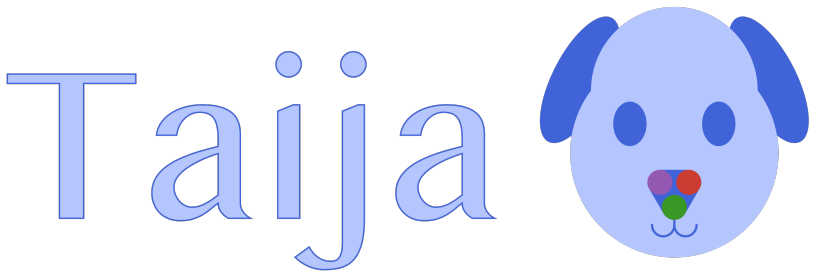



Figure 16: Trustworthy AI in Julia: github.com/JuliaTrustworthyAI

# References

Altmeyer, Patrick, Giovan Angela, Aleksander Buszydlik, Karol Dobiczek, Arie van Deursen, and Cynthia C. S. Liem. 2023. "Endogenous Macrodynamics in Algorithmic Recourse." *2023 IEEE Conference on Secure and Trustworthy Machine Learning (SaTML)*, 418–31. https://doi.org/10.1109/satml54575.2023.00036.

Altmeyer, Patrick, Andrew M Demetriou, Antony Bartlett, and Cynthia C. S. Liem. 2024. "Position: Stop Making Unscientific AGI Performance Claims." *International Conference on Machine Learning*, 1222–42. https://proceedings.mlr.press/v235/altmeyer24a.html.

Altmeyer, Patrick, Arie van Deursen, and Cynthia C. S. Liem. 2023. "Explaining Black-Box Models through Counterfactuals." *Proceedings of the JuliaCon Conferences* 1: 130. https://doi.org/10.21105/jcon.00130.